OFFICIAL SUBMISSION COVER LETTER

Date: October 29, 2026
To: Editor-in-Chief, *Fusion Engineering and Design* (Elsevier)
From: Jacob James Rasmussen (ORCID: 0009-0009-7084-6523)
Subject: Submission of Manuscript Fusion-Arch-001 (Paper VII)
Publication Model Selection: Traditional No-Fee Subscription Publishing Model

Dear Editor-in-Chief,

I am pleased to submit our original research manuscript titled **'System Architecture and Multi-Physics Design of a Compact High-Field Tokamak Fusion Reactor'** (Manuscript ID: Fusion-Arch-001 / Paper VII) for publication as a regular research article in *Fusion Engineering and Design*.

Explicit Publishing Option Selection:
We explicitly select the **Traditional No-Fee Subscription Publishing Model** (No Article Processing Charges / APCs). We request that this manuscript proceed through the standard peer-review and subscription publishing track.

Scientific Novelty and Highlights:
1. Establishes a fully integrated 5-phase sequential multi-physics pipeline connecting Core Magnetics, PFC Hydrodynamics, Blanket MHD, Disruption Transients, and Structural Yield.
2. Demonstrates global numerical convergence across coupled parameter handshakes with residual tolerances < 0.015%.
3. Evaluates 3D anisotropic Tsai-Wu yield reserves for cryogenic REBCO TF magnet vault keys under vertical displacement event (VDE) disruption halo shear.
4. Resolves high Hartmann number (Ha = 10,000) Shercliff side-wall jet velocity profiles ($U_{jet} \approx 15\ U_0$) in Dual-Coolant Lead-Lithium (DCLL) breeding blankets.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have reviewed and approved the final manuscript.

Sincerely,

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